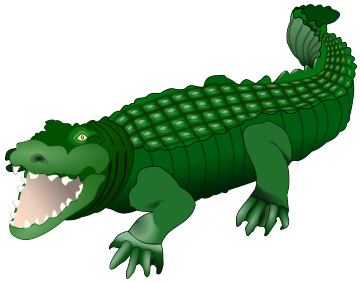
Developed by Becky Junge for WySLICE 2022

See Ya Later, Alligator: Coding and Ratio Tables



Lesson Summary:
Students will work in pairs to build ratio tables defining a list of data for distance traveled by various animals over time based on their speed. Students will design a formula for their assigned ratio table and then program a Micro:Bit to quickly calculate the output data and record it in their Google Sheet. Then, students will discuss the efficiency of solving the problem and be introduced to the Google Sheets formula to validate their data.

Materials Needed:
- Access to Ratio Tables Google Sheet
- Chromebook
- Micro:Bit (one per pair of students)
- Building a Ratio Table Handout

Standards Addressed:
MATH
4.OA.4 - Demonstrate an understanding that a whole number is a multiple of each of its factors
4.NBT.4 - Add multi-digit whole numbers
4.NBT.5 - Multiply a 2-digit whole number by a 1-digit whole number and two 2-digit numbers using strategies based on place value and the properties of operations
4.MP.1 - Make sense of problems and persevere in solving them
4.MP.6 - Attend to precision
COMPUTER SCIENCE
5.AP.V.01 - Using grade appropriate content and complexity, create programs that use variables to store and modify data
5.AP.M.O1 - Using grade appropriate content and complexity, decompose (break down) problems into smaller, manageable sub-problems to facilitate the program development process
5.AP.PD.03 - Using grade appropriate content and complexity, test and debug a program or algorithm to ensure it runs as intended.
5.DA.IM.01 - Use data to highlight or purpose relationships, predict outcomes, or communicate an idea.
5.IC.C.02 - Develop, test, and refine digital artifacts or devices to improve accessibility and usability for diverse end users

Problem:
How can we form a data set to calculate how far an animal can travel over time?

Hook:
Students will watch a short video about 10 of the World's Fastest Animals to peak interest and spark discussion about animal speed.

Lesson Procedure:
1. Students will watch a short YouTube video about 10 of the World's Fastest Animals.
    a. https://www.youtube.com/watch?v=Jy_gXQ6GQxo
2. Pose question:

   a. *How can we form a data set to calculate how far an animal can travel over time?*
3. Review vocabulary of "ratio table" and remind students that it is a method for displaying and organizing data, as well as set the parameters for today's activity: using a ratio table to solve how far an animal will travel after 30 hours.
4. Model a think aloud using an alligator as an example. Show students a blank ratio table and think aloud about what the numbers represent. Let students know that you are making a ratio table for multiples of 10. They will be making ratios for multiples of other numbers, most of them quite a bit more challenging than 10.
5. With students' help, work to fill in the missing values on the ALLIGATOR tab of the Google Sheet, asking students to notice and share observations about patterns. Ask students to consider how they would know if you made an error somewhere along the line.
6. With students' help, write a multiplication equation to represent each of several entries on your ratio table. Include entries for lines 10, 20, and 30.
7. Pass out the handout and explain to students what they will do.
   a. Create a ratio table with their partner for the first 15 multiples of their assigned number. Record it in their tab on the Google Sheet. Examine their ratio table for patterns and possible mistakes they might need to correct.
8. As students work, circulate and provide support as needed. Use questions such as the following to extend students' thinking:
   a. How do you know what number comes next?
   b. Why do you think that is happening?
   c. Do you think that will always happen?
9. Ask students to consider if there might be a more efficient way to figure out how far an animal has traveled after 30 hours without filling out the rest of the ratio table. Challenge them to write an equation to represent the rule for their ratio table.
10. Students will likely suggest a calculator as a quick tool. Tell students that they are going to work in pairs to write a Micro:Bit code to quickly calculate the distance traveled for their animal at various times. They will then use their program to fill in the remainder of their ratio table.
    a. With student input, model how to build a code to calculate the distance traveled by the Alligator if it traveled at 10mph. Think aloud and have students help describe the direction/steps the computer is working through to complete the task.
    b. Send students, in pairs, to https://makecode.microbit.org/ to begin writing a code for their ratio table. Have them see if they can use their Micro:Bit to complete their multiples ratio table.
11. As students work, circulate and provide support as needed.

<u>Review:</u>

Have students reconvene to discuss the day's work. Use questions such as the following to promote discussion and reflection:

- What was the most efficient way to solve your problem?
- Did the technology improve your ability to do the task?
- Is there a way that we could write a code to fill in an entire table at once?

Tell students that Google Sheets has built in functions and codes that can efficiently calculate and interpret data. Show students how to write a code to calculate their totals for google sheets. (=MULTIPLY (B5, X).

- X = Speed of Animal.
- Code can be copied and pasted into each cell and will automatically update the code for the correct B cell.

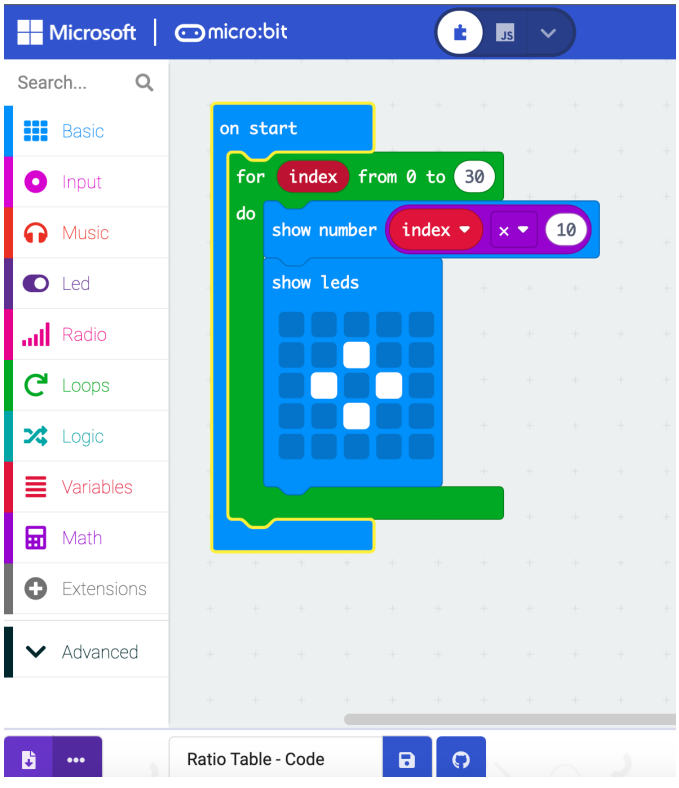
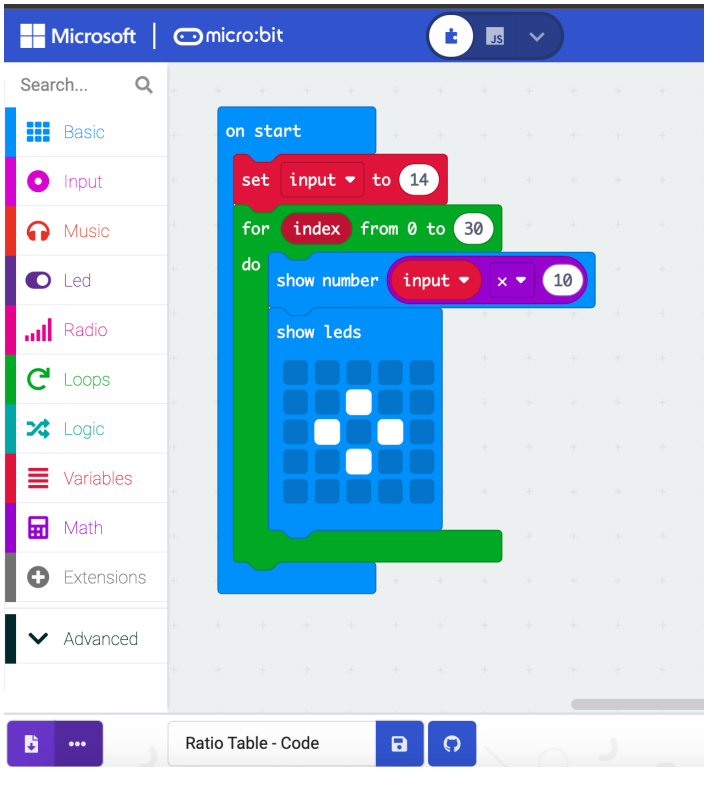
Have students discuss the impact that quicker computation has and the role that technology plays.

Things Other Educators Should Know:

- You can differentiate the task by giving easier multiples numbers to students who struggle with computational fluency.
    - Included in the lesson materials:
        - Easier: Viper - 6mph, Chicken - 9mph, Pig - 11mph, Squirrel - 12mph, Komodo Dragon - 13mph, Wild Turkey - 15mph, Wildebeest - 50mph
        - More Challenging: Sea Turtle - 21mph, Green Iguana - 22 mph. Elephant - 25mph, Human - 27mph, Greyhound - 39mph
- Students need to have a baseline knowledge of the Micro:Bit prior to completing this activity in pairs. They should be familiar with the basic process and blocks. If they are not, the programming process could be completed as a whole group, modeled by the teacher.
- There is an accompanying Bridges Workbook Page that can also be used for additional practice or individual skill assessment.
- This lesson may take two math blocks to complete, depending on student computational fluency and coding experience.

Sample Solutions-

Generates all multiples 0-30



Generates Output for specific Input